Order-Memory Compression of Zero-Repair Trees: An Adaptive-Readout Fix and a Growing-Gap Warning

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Abstract

We report on a family of solvers that approximate the finite-horizon zero-repair value $\mathscr{D}_r(B_8)$ by restricting to trees whose node law depends on history only through a signature $\sigma_K(h) = (\mathrm{count}(h), \mathrm{suffix}_K(h))$ – the branch-count vector together with the last K symbols. This interpolates between the fully count-collapsed class ($K = 0$, previously shown to be substantially lossy) and the exact problem ($K = r - 1$). A fixed top-4-of-5 readout in the one-shot binary32 field solver used for this class fails (*certified infeasible*) from depth 5 onward, independent of K. We implement and verify an adaptive margin-based readout that resolves this at $r = 5, K = 3$, and we show, by an explicit counterexample, that the heuristic solver's own successful output at that setting violates the class-inclusion hierarchy $\mathscr{D}_r^K(B_8) \geq \mathscr{D}_r^{K+1}(B_8)$ – a valid, exactified, verified tree can still be a materially suboptimal witness within its own class. Direct exact optimization within the K-restricted class (bypassing the field and readout entirely) gives certified values consistent with the hierarchy. The resulting data give the main finding of this note: the gap between $\mathscr{D}_r^K(B_8)$ and the best available reference, at *fixed* $K \in \{1, 2\}$, does not plateau as r grows from 2 to 5 – it grows, and the available data are consistent with accelerating growth. This is evidence against fixed-memory order compression as a route to a tractable exact solver, and reframes the target question as one about the growth rate $K(r)$ required to keep the gap controlled.

1 Setting

Fix the quartic pair P, Q on $\{0, \dots, 4\}$, support cutoff $M = 8$, and the certified three-atom target B_8 used throughout this project. For a history $h \in \{0, \dots, 4\}^{\leq r}$, write

$$\sigma_K(h) = \big(\mathrm{count}(h),\ \mathrm{suffix}_K(h)\big),$$

where $\mathrm{count}(h) \in \mathbb{Z}^5_{\geq 0}$ records how many times each symbol occurs in h, and $\mathrm{suffix}_K(h)$ records the last $\min(K, |h|)$ symbols of h, in order. A depth-r *K-memory tree* is a zero-repair tree in which every node's law is required to depend on its history only through σ_K. Let $\mathscr{D}_r^K(B_8)$ denote the minimum root mean over such trees whose leaves stochastically dominate B_8.

Proposition 1 (Monotone hierarchy). *For every r and every $0 \leq K \leq K' \leq r - 1$,*

$$\mathscr{D}_r(B_8) = \mathscr{D}_r^{r-1}(B_8) \leq \mathscr{D}_r^{K'}(B_8) \leq \mathscr{D}_r^K(B_8).$$

Proof. Every K-memory tree is in particular a K'-memory tree for $K' \geq K$ (a function that depends on history only through σ_K also depends on it only through the finer signature $\sigma_{K'}$, simply ignoring the extra information). Hence the K-memory feasible class is contained in the K'-memory feasible class, and the minimum over a subset is at least the minimum over the larger set. At $K = r - 1$, $\sigma_{r-1}(h)$ recovers h exactly (up to the redundant count information), so the restriction vanishes and $\mathscr{D}_r^{r-1}(B_8) = \mathscr{D}_r(B_8)$. □

This hierarchy is not a numerical observation; it is definitional, and it is the yardstick against which every solver output below is checked.

2 A readout failure independent of K

The one-shot solver architecture used throughout this project – one deterministic binary32 field, one frozen support readout, one exactification LP with no add-back – was previously found to fail (certified infeasible) on the unrestricted problem at depth 6. The same failure mode reappears inside the K-memory class at smaller depths: a fixed top-4-of-5 readout produced a certified-infeasible frozen face at $r = 5$ for $K = 3$, and at other (r, K) pairs at comparable depth.

2.1 Adaptive margin readout

We implement a second readout policy. For each semantic row (a node together with a carry state j and fresh symbol y) with all five branch values admissible, compare the fourth- and fifth-largest field activations. If the margin between them is at or above a threshold τ, the row is compressed to its top four values, exactly as before. If the margin is below τ, all five values are retained for that row only. This is still a single deterministic pass over a frozen field output – no field feedback, no iteration, no second field run – differing from the original policy only in *which* atoms one readout keeps.

Observation 1 (The fix resolves the specific failure)**.** *At $r = 5$, $K = 3$, $\tau = 0.001$: 15,620 rows are trivial (five or fewer admissible branches to begin with), 18,777 rows are compressed under the margin test, and 748 rows are kept in full because their margin fell below threshold. The resulting frozen face is feasible; exactification succeeds and independently verifies (freshness, branch-marginal, child-generation, and leaf-dominance errors all at machine precision).*

3 A verification-discipline failure, caught by the hierarchy itself

The adaptive-margin fix, run at $r = 5$, $K = 3$, $\tau = 0.001$, returned a verified exact tree with root mean

$$0.791644578279324.$$

This passed every check internal to that single run: normalization, freshness, branch marginals, child generation, and leaf dominance were all at machine precision.

Observation 2 (Hierarchy violation)**.** $K = 1$ *is a strictly poorer memory class than* $K = 3$ *(Proposition 1), so* $\mathscr{D}_5^{K=3}(B_8) \leq \mathscr{D}_5^{K=1}(B_8)$ *must hold. The independently computed exact value* $\mathscr{D}_5^{K=1}(B_8) = 0.700727081277764$ *is smaller than* 0.791644578279324*. The field-plus-readout output is therefore* not $\mathscr{D}_5^{K=3}(B_8)$*: it is a valid but substantially suboptimal point in the* $K = 3$ *class.*

This is the central methodological point of this note. A successful exactification with a passing internal verifier certifies that the returned object is a genuine feasible tree in its class. It does not certify optimality, and nothing internal to the run detects the gap: only a cross-check against an independently established bound in the same hierarchy exposed it. Every number in Section 4 below was obtained by discarding the field and readout entirely and solving the K-restricted class directly, so that the reported values are optimal within their class, not merely feasible.

4 Direct exact optimization within the K-restricted class

Bypassing the field and readout, we solve the full linear program over the σ_K-signature DAG directly, with no support restriction beyond the one already implied by K itself.

Both values are consistent with the hierarchy ($0.6233 \leq 0.6513 \leq \mathscr{D}_5^{K=1}(B_8) = 0.7007$). The solve time more than doubled from $K = 2$ to $K = 3$ at fixed $r = 5$ ($781.3/367.2 = 2.13$), on top

r	K	$\mathscr{D}_r^K(B_8)$	solve time
5	2	0.651329712088707	367.2 s
5	3	0.623322004930434	781.3 s

of an already large absolute cost; this class is not becoming cheap to solve exactly as K grows, even though its state space grows only polynomially in r for fixed K.

5 The growing-gap finding

Define, for the best currently available reference $\widehat{\mathscr{D}}_r(B_8)$ at depth r (the exact value where known, or the tightest verified upper bound otherwise) the residual

$$\operatorname{gap}(r, K) := \mathscr{D}_r^K(B_8) - \widehat{\mathscr{D}}_r(B_8) \geq 0.$$

At $r = 5$ the only available reference is $\mathscr{D}_5^{K=3}(B_8)$ itself, so the reported values are lower bounds on the true gap against $\mathscr{D}_5(B_8)$, which is itself unknown and can only be smaller.

r	$\operatorname{gap}(r,1)$	$\operatorname{gap}(r,2)$
2	0	–
3	0.0186431427	0
4	0.0514670626	0.0147072406
5	≥ 0.0774050763	≥ 0.0280077072

Observation 3 (No plateau)**.** *At both $K = 1$ and $K = 2$, $\operatorname{gap}(r, K)$ increases at every step for which data exist, and the increase does not shrink: the $K = 2$ gap at least nearly doubles from $r = 4$ to $r = 5$ ($\geq 1.90\times$), and the $K = 1$ gap grows by a factor of at least $1.50\times$ over the same step. Because the $r = 5$ entries are lower bounds, the true growth factors may be larger still.*

5.1 What this does and does not show

Four data points at $K = 1$ and three at $K = 2$ do not establish an asymptotic law. In particular:

- We cannot rule out that $\operatorname{gap}(r, K)$ eventually plateaus for $r > 5$; the observed growth could be a transient of small r.
- We cannot rule out a faster-than-geometric (e.g. genuinely unbounded, non-decaying-increment) growth either; the data are equally compatible with several qualitatively different regimes at this sample size.
- The $r = 5$ numbers are contaminated by using $\mathscr{D}_5^{K=3}(B_8)$, itself only an upper bound, as the reference; every reported $r = 5$ gap is a *lower* bound on the true gap against $\mathscr{D}_5(B_8)$.

What the data do support, without needing to resolve the rate, is a direct warning against the working hypothesis that motivated this whole line of attack: that a small, depth-independent memory K might suffice to approximate $\mathscr{D}_r(B_8)$ well for all r. At both memory lengths tested, the approximation is getting *worse*, not better, exactly over the range where it was hoped to stabilize.

6 Relation to the dyadic doubling-premium question

This note's object, $\mathscr{D}_r^K(B_8)$, holds the target law B_8 *fixed* and asks how a memory-restricted class approximates the value at growing depth r. This is a different question from Problem 8 of the latent-fibers commentary, which asks about a prefix-compatible *sequence* of targets B_{2^k} and a bounded increment $\mathscr{D}_{2^{k+1}}(B_{2^k}) - \mathbb{E}B_{2^k}$ across dyadic scales. Growth in $\mathrm{gap}(r, K)$ at fixed B_8 says nothing directly about whether a cleverly chosen dyadic target sequence admits a bounded-memory or bounded-premium construction; the two should not be conflated. This note's contribution is confined to the fixed-target compressibility question.

7 An open problem

Problem 14 (Required memory growth rate)**.** *Does there exist a function $K(r) = o(r)$ such that $\mathrm{gap}(r, K(r)) \to 0$ as $r \to \infty$? If so, what is the slowest such rate – is $K(r) = O(\log r)$ attainable, or is a stronger rate required? Equivalently: is there a genuine exponential compression of the history signature that keeps $\mathscr{D}_r^{K(r)}(B_8)$ asymptotically exact, or does the growing-gap trend in Section 5 continue in a way that forces $K(r)$ to be a non-trivial fraction of r?*

8 Scope

This note reports one engineering fix (adaptive margin readout, verified to resolve one specific certified infeasibility), one methodological finding (a heuristic solver's successful, internally-verified output can still violate an external monotonicity constraint of its own problem class, and only checking against that constraint exposes it), and one empirical warning (the fixed-K approximation gap grows, not shrinks, over the only range tested). It does not establish $\mathscr{D}_r(B_8)$ at $r = 5$ or beyond, does not determine the asymptotic behavior of $\mathscr{D}_r^K(B_8)$ for any fixed K, and does not bear directly on Problem 8 or on $B_n = \Theta(\log n)$. Code implementing the adaptive readout and the direct exact K-class solver accompanies this note.